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Project Proposal

Introduction:

Evaluating young talent is a pivotal, but also incredibly difficult part of building a successful Major League Baseball (MLB) organization. Prospects vary significantly in age, position, developmental background, and access to resources, making it difficult to accurately predict their future performance. Sims & Addona (2016) investigated how age influences MLB draft selections and subsequent performance, finding evidence that younger high school players were systematically undervalued. This raises questions about similar bias in prospect rankings. Gentile (2022) examined MLB's academy system in the Dominican Republic, highlighting structural inequalities in the treatment and development of international prospects. These findings are important because players entering professional baseball through international signing markets may have different experiences and opportunities than players entering through the domestic draft raising questions about whether prospect evaluations are equally accurate across players from different developmental backgrounds.

Furthermore, modern prospect rankings rely on a combination of subjective scouting grades and objective stats and metrics (e.g., height, weight, max exit velocity). As the industry shifts towards valuing these objective metrics over functional game skills it becomes crucial to evaluate whether prospect rankings suffer from survivorship bias, overvaluing specific physical archetypes or raw metrics while marginalizing unconventional players.

Research Questions:

1. Systemic Background Biases: Do FanGraphs' evaluations systematically misvalue players based on demographic factors like age and developmental origins (highschool, college, or international signing), and to what extent do algorithmic "Risk" penalties reflect these differences? How is a player's signing bonus related to their developmental origin and what impact does that have on their evaluation?

2. Biometric Archetypes: Do FanGraphs' evaluations exhibit systemic bias toward certain physical archetypes (e.g., height, weight), and do players who fall outside these traditional molds systematically outperform their projections?

3. "Stuff" vs. "Feel" Valuation: Do FanGraphs' evaluations overvalue raw, measurable tools (e.g., fastball velocity, max exit velocity) at the expense of functional game skills (e.g., command, bat control, hit tool)?

Motivation:

Jack, Dash, and Miller are all baseball players who closely follow the MLB. While signing top free agents and trading for star players can be very advantageous, MLB organizations must also identify and develop young talent. FanGraphs' The Board tries to evaluate these players by ranking prospects by their projected future value, but many highly ranked prospects fail to have a successful career in the MLB, and many lower-ranked prospects greatly outperform their evaluations and become impact players. This made us wonder how accurately these rankings predict future MLB performance, and whether certain groups of players are systematically overestimated or underestimated based on characteristics such as age, signing market, background, and physical tools. By examining where the predictions are strongest and weakest, we hope to better understand potential biases that could influence opportunities for different groups of players. While previous research examined how factors like age and

developmental background influence baseball scouting and player development, we hope to investigate whether these differences are reflected in publicly available prospect evaluations.

Data:

We plan to create a dataset using FanGraphs' The Board prospect evaluations from 2020 through 2023. We will include all available prospects from each year, instead of just the top 100. The data includes general characteristics like each player's name, evaluation year, prospect ranking, Future Value (FV) rating, risk factor, age, position, MLB organization, where they signed from, and their singing bonus. Scouting information includes height, weight, exit velocity, and current/projected tool grades on the 20–80 scale (e.g., 40/45 power). We will combine these evaluations with subsequent MLB performance data, including whether a player reached MLB, games played, and Wins Above Replacement (WAR). WAR will serve as our primary measure of performance because it allows comparisons between pitchers and position players. We will clean and standardize player names and positions to match players across datasets and account for players who appear on The Board in multiple years. Since some prospects may not have had sufficient time to establish MLB careers, we will use a consistent follow-up period after each evaluation to make comparisons more equitable.

Methods:

We will begin with exploratory data analysis using visualizations such as scatterplots, histograms, and boxplots to investigate relationships between the variables in our research questions such as FV ratings, prospect rankings, age, signing market, signing bonus, background and MLB performance (WAR).

For our first research question, we will use multiple linear regression to examine whether FV, ranking, age, developmental origin, and signing bonuses predict WAR. If age, developmental

origin, and signing bonuses are statistically significant predictors it indicates that FV fails to fully capture their impact. We will use logistic regression testing to determine which groups are most likely to be named high risk and whether this is actually reflected in their performance.

For our second question, we will see whether the residuals of our model predicting WAR from FV and prospect rankings are related to players' physical dimensions. For example, if shorter and lighter players are correlated with positive residuals that would mean they are undervalued.

For the third question we will separate the scouting information in the data set into “Stuff” (metrics like exit velocity and hard-hit percentage) and “Feel” variables (scouting grades). Once again using MLR we will see which of these variables FV ratings value more and then see which of those actually best predict WAR.

Our methods may evolve as we learn additional techniques, potentially including random forests or deep learning to identify biometric archetypes.

Research Expectations:

We expect higher-FV prospects to achieve greater MLB success, although considerable variation is likely. Based on previous research, we expect that age and background may influence evaluation accuracy, with certain younger players signed from US high schools or internationally being undervalued. We expect that international players will receive disproportionately higher “Risk” grades compared to domestic counterparts but may outperform their FV projections, suggesting the algorithm's risk penalties somewhat compensate for developmental uncertainty. We also hypothesize a strong "sunk cost" bias, where higher signing bonuses and draft position inflate FV ratings even after controlling for their underlying tools and performance. As for biometric archetypes, we expect to find that the ranking algorithm exhibits a strong preference

for traditional physical molds (e.g., pitchers 6'3" or taller). Players outside these traditional archetypes will then be undervalued and underrepresented in these rankings and likely generate higher than predicted WAR. We also predict that FV ratings will heavily value "Stuff" variables while “Feel” variables will translate disproportionately well to WAR.

Bibliography:

Gentile, Patrick C. “MLB’s Neocolonial Practices in the Dominican Republic Academy System.” *Journal of Sport and Social Issues*, vol. 46, no. 3, 2022, pp. 269–292. https://journals.sagepub.com/doi/10.1177/0193723521991404

Sims, Justin, and Vittorio Addona. “Hurdle Models and Age Effects in the Major League Baseball Draft.” *Journal of Sports Economics*, vol. 17, no. 7, 2016, pp. 672–687. https://journals.sagepub.com/doi/10.1177/1527002514539516

Work Agreement:

We’ve all agreed to contribute equally in terms of quantity and quality. For bigger sections and decisions, we will all work together. We also have a groupchat, and for smaller work, we can do work individually, and check with both other group members for approval. We will all do the code and cleaning together in order to make sure each group member understands the methodology and processes used. We also plan to complete all work ahead of deadlines in order to give the other members time to verify and approve of the work. Before edits and changes are made, we will notify other group members and get approval before putting them into place.